Kubernetes APM & Infrastructure Runbooks

RAG KNOWLEDGE BASE & OPERATIONAL TROUBLESHOOTING REFERENCE (50 ISSUES)

50	12	12	26
TOTAL RUNBOOKS	POD & CONTAINER	NODE & INFRA	CONTROL & APM

SECTION 1: POD & CONTAINER LEVEL ISSUES (1 – 5)

1. KubePodCrashLooping — POD-001

Symptoms: High restart count `kube_pod_container_status_restarts_total`; status `CrashLoopBackOff`.
Impact: Application service degradation or localized module unavailability.
Diagnosis:
1. Fetch previous container logs: `kubectl logs <pod> -n <ns> --previous`
2. Inspect termination reasons: `kubectl describe pod <pod> -n <ns>`

Resolution: Fix application runtime errors, missing environment variables, or entry point scripts.

2. OOMKilled (Exit Code 137) — POD-002

Symptoms: Container exit code 137; `container_memory_working_set_bytes` equals limit.
Impact: Sudden termination of active worker processes and in-flight transactions.
Diagnosis:
1. Inspect memory usage trends prior to crash in Prometheus/Grafana.
2. Run `kubectl describe pod <pod>` and look for `Last State: Terminated (OOMKilled)`.

Resolution: Fix memory leaks in code or increase container memory limits in the manifest.

3. ImagePullBackOff / ErrImagePull — POD-003

Symptoms: Pod stuck in `ImagePullBackOff` state; waiting reason flag active.
Impact: Pod cannot start; new code deployments or bug fixes fail to deploy.
Diagnosis: Run `kubectl describe pod <pod>` and check Events for registry auth or 404 errors.
Resolution: Correct image tag typos or update `imagePullSecrets` in service account.

4. PodPending / Unscheduled Pod — POD-004

Symptoms: `kube_pod_status_phase{phase="Pending"} > 0`.
Impact: Applications fail to scale up to meet workload demand.
Diagnosis: Check describe output for scheduler warnings (Insufficient CPU/Memory, Taints).
Resolution: Add worker nodes, lower pod resource requests, or adjust node selectors.

5. CreateContainerConfigError — POD-005

Symptoms: Pod status displays `CreateContainerConfigError`.
Impact: Container initialization halts immediately.
Diagnosis: Check describe pod events for missing ConfigMap or Secret keys.
Resolution: Create missing ConfigMap/Secret objects or align key names in manifest.

SECTION 1: POD & CONTAINER LEVEL ISSUES (6 – 12)

6. CPU Throttling Alert

POD-006

Symptoms: High ratio of `container_cpu_cfs_throttled_periods_total`.
Impact: Severe request latency spikes without high reported CPU utilization.
Diagnosis: Check if pod CPU usage matches its configured limit during GC or thread bursts.
Resolution: Increase CPU limits, remove strict CPU limits (use requests), or tune concurrency.

7. Pod Liveness Probe Failure

POD-007

Symptoms: Repeated restarts with event `Liveness probe failed`.
Impact: Healthy containers killed prematurely due to transient slow responses.
Diagnosis: Test liveness HTTP endpoint response times under application load.
Resolution: Increase `initialDelaySeconds`, `timeoutSeconds`, or `failureThreshold`.

8. Pod Readiness Probe Failure

POD-008

Symptoms: Pod marked `0/1 READY`; removed from Service endpoints.
Impact: Reduced active capacity to process incoming application traffic.
Diagnosis: Curl readiness path locally: `kubectl exec <pod> -- curl localhost:<port>/healthz`.
Resolution: Restore downstream database connectivity or tune readiness probe parameters.

9. Pod Stuck in Terminating

POD-009

Symptoms: Pod stays in `Terminating` state past grace period.
Impact: Locks attached storage volumes and prevents rescheduled instances from starting.
Diagnosis: Check for blocking finalizers in pod YAML or hung node CSI drivers.
Resolution: Remove stuck finalizers or force delete: `kubectl delete pod <p> --force --grace-period=0`.

10. Ephemeral Storage Limit Exceeded

POD-010

Symptoms: Evicted with message `ephemeral local storage usage exceeds limit`.
Impact: Pod eviction and loss of temporary files stored on local rootfs.
Diagnosis: Check for runaway local log files or temp files written inside container.
Resolution: Route logs to stdout, implement log rotation, or mount emptyDir volumes.

11. CreateContainerError

POD-011

Symptoms: Pod status displays `CreateContainerError`.
Impact: Container fails to launch entirely.
Diagnosis: Check describe logs for mount collisions or host path permission errors.
Resolution: Fix storage driver volume mounts or update security context permissions.

12. High Pod Eviction Rate

POD-012

Symptoms: Surge in `kube_pod_status_reason{reason="Evicted"}`.
Impact: Disruption across multiple service instances.
Diagnosis: Inspect node conditions for Memory, Disk, or PID pressure.
Resolution: Add cluster node capacity or prune unused container images and volumes.

SECTION 2: NODE & CLUSTER INFRASTRUCTURE ISSUES (13 – 20)

13. KubeNodeNotReady — INF-013

Symptoms: Node condition `Ready == 0` or status shows `NotReady`.
Impact: Node workloads evicted; total cluster compute capacity drops.
Diagnosis: SSH to node and verify status of kubelet: `systemctl status kubelet`.
Resolution: Restart kubelet/containerd runtime or replace underlying cloud VM instance.

14. Node Memory Pressure — INF-014

Symptoms: Condition `MemoryPressure == 1` active on node.
Impact: Kubelet triggers evictions of non-Guaranteed QoS pods.
Diagnosis: Run `kubectl top nodes` and identify highest memory consuming pods.
Resolution: Cordon/drain node, set resource limits on pods, or scale node pool.

15. Node Disk Pressure — INF-015

Symptoms: Condition `DiskPressure == 1` active on node.
Impact: Pod scheduling halted on node; image garbage collection triggered.
Diagnosis: Check disk usage on node host rootfs and container storage directory (`df -h`).
Resolution: Prune unused images (`crictl rmi --prune`) or expand node storage volume.

16. Node PID Pressure — INF-016

Symptoms: Condition `PIDPressure == 1` active on node.
Impact: Fork failures inside containers; node cannot start new processes.
Diagnosis: Check process count per pod and host PID ceiling (`cat /proc/sys/kernel/pid_max`).
Resolution: Enforce `podPidsLimit` in Kubelet config and fix process/thread leaks in app code.

17. KubeletTooManyPods — INF-017

Symptoms: `kubelet_running_pods` reaches configured max capacity (default 110).
Impact: New pods fail to schedule on node despite available CPU/memory.
Diagnosis: Count running pods on target node via kubectl field-selector.
Resolution: Raise maxPods parameter in Kubelet configuration or add more worker nodes.

18. KubeCPUOvercommit — INF-018

Symptoms: Sum of CPU requests exceeds allocatable node CPU capacity.
Impact: Risk of CPU starvation during load spikes; reduced tolerance to node loss.
Diagnosis: Aggregate `kube_pod_container_resource_requests` vs allocatable metrics.
Resolution: Right-size pod CPU requests or enable Cluster Autoscaler.

19. KubeMemoryOvercommit — INF-019

Symptoms: Sum of memory requests exceeds total allocatable cluster memory.
Impact: Cascading OOM evictions across cluster if a single node fails.
Diagnosis: Compare total requested memory against allocatable capacity across node pools.
Resolution: Adjust memory requests down based on actual usage or provision extra nodes.

20. Kubelet System OOM — INF-020

Symptoms: Kubelet daemon killed by Linux kernel OOM killer.
Impact: Node status drops to `NotReady`; control plane loses node management.

SECTION 2: NODE & CLUSTER INFRA (21 – 24)

21. Node Clock Drift — INF-021

Symptoms: Clock skew detected (`node_timex_offset_seconds > threshold`).
Impact: TLS handshake failures, auth token expiration issues, broken APM traces.
Diagnosis: Run `timedatectl status` or `chronyc tracking` on node host.
Resolution: Enable and configure `chrony` or `systemd-timesyncd` on worker nodes.

22. High Node Network Drop Rate — INF-022

Symptoms: Spikes in `node_network_receive_drop_total` .
Impact: Network latency, TCP retransmissions, dropped APM trace spans.
Diagnosis: Check NIC hardware errors via `ethtool -S` and kernel buffer limits.
Resolution: Increase NIC ring buffer size (`ethtool -G`) and tune network sysctl settings.

23. Host Conntrack Table Full — INF-023

Symptoms: Kernel error `nf_conntrack: table full, dropping packet` .
Impact: Random TCP connection drops across all pods hosted on the node.
Diagnosis: Read active conntrack count: `cat /proc/sys/net/netfilter/nf_conntrack_count` .
Resolution: Increase `sysctl net.netfilter.nf_conntrack_max` value.

24. Unbound PersistentVolumeClaim (PVC) — INF-024

Symptoms: `kube_persistentvolumeclaim_status_phase{phase="Pending"} == 1` .
Impact: StatefulSet or Deployment pods locked in `Pending` state.
Diagnosis: Inspect PVC details (`kubectl describe pvc <pvc>`) for provisioner errors.
Resolution: Fix storage driver permissions, StorageClass definition, or quota.

SECTION 3: CONTROL PLANE & CORE SERVICES (25 – 28)

25. KubeAPILatencyHigh — CTL-025

Symptoms: P99 API latency > 1s (`apiserver_request_duration_seconds`).
Impact: Slow kubectl commands, delayed autoscaling, controller timeouts.
Diagnosis: Identify slow API verbs in Prometheus and check etcd disk sync times.
Resolution: Optimize noisy list calls, scale API server instances, or use faster SSDs.

26. etcdNoLeader — CTL-026

Symptoms: `etcd_server_has_leader == 0` .
Impact: CRITICAL: Cluster state writes blocked; API server fails or goes read-only.
Diagnosis: Run `etcdctl endpoint health` and inspect etcd logs for network partitions.
Resolution: Resolve master network partitions or restore etcd cluster from backup.

27. High etcd Disk Commit Latency — CTL-027

Symptoms: `etcd_disk_wal_fsync_duration_seconds` p99 > 10ms.
Impact: API server slowness, etcd leader elections, cluster instability.
Diagnosis: Measure disk IOPS and write latency on etcd nodes using `iostat -x 1` .
Resolution: Provision dedicated low-latency NVMe storage for etcd data.

28. KubeControllerManagerDown — CTL-028

SECTION 3: CONTROL PLANE & CORE SERVICES (29 – 34)

29. KubeSchedulerDown — CTL-029

Symptoms: `up{job="kube-scheduler"} == 0`.
Impact: Unscheduled pods stay permanently in Pending state.
Diagnosis: Inspect kube-scheduler static pod logs on master node.
Resolution: Correct startup flags/config syntax and restart scheduler container.

30. CoreDNS Performance Bottleneck — CTL-030

Symptoms: High CoreDNS latency or dropped DNS requests.
Impact: Inter-service timeouts, slow APM trace generation, connectivity failures.
Diagnosis: Check CoreDNS CPU usage and test lookups with dig/nslookup in container.
Resolution: Autoscale CoreDNS pods and deploy NodeLocal DNSCache.

31. Kube-Proxy Sync Failure — CTL-031

Symptoms: `kubeproxy_sync_proxy_rules_duration_seconds` failing.
Impact: Stale routing rules; traffic sent to dead pods or dropped.
Diagnosis: Check kube-proxy pod logs and iptables rule counts on worker node.
Resolution: Migrate proxy mode from iptables to ipvs or eBPF (Cilium).

32. Control Plane Cert Expiration — CTL-032

Symptoms: Alert `KubeClientCertificateExpiration` active (< 7 days).
Impact: Control plane operations and worker-to-master communications fail completely.
Diagnosis: Verify expiration dates: `kubeadm certs check-expiration`.
Resolution: Renew certificates manually: `kubeadm certs renew all`.

33. KubeQuotaExceeded — CTL-033

Symptoms: Resource quota usage ratio reaches 100%.
Impact: Cannot deploy new workloads or services in targeted namespace.
Diagnosis: Run `kubectl describe resourcequota -n <ns>`.
Resolution: Clean up unused resources or increase ResourceQuota limits.

34. Deployment Generation Mismatch — CTL-034

Symptoms: `observed_generation != metadata_generation`.
Impact: Deployment controller fails to reconcile desired update.
Diagnosis: Check deployment events for RBAC or invalid field errors.
Resolution: Correct deployment spec syntax or resolve RBAC permissions.

SECTION 4: APM & MICROSERVICE METRICS (35 – 36)

35. HTTP 5xx Server Error Rate Spike — APM-035

Symptoms: APM metric `http_requests_total{status=~"5.."}` spikes rapidly.
Impact: Direct user-facing application failures and degraded SLA.
Diagnosis: Filter distributed traces by status 500 to locate failing endpoint code.
Resolution: Roll back recent deployment or issue hotfix for unhandled runtime exception.

36. High P99 Transaction Latency Spike — APM-036

SECTION 4: APM & MICROSERVICE METRICS (37 – 44)

37. DB Connection Pool Exhaustion — APM-037

Symptoms: `db_pool_wait_time_ms` spikes; active connections reach limit.
Diagnosis & Fix: Check DB server max connections. Increase pool max size and ensure connections release in `finally` blocks.

38. JVM Garbage Collection Pauses — APM-038

Symptoms: `jvm_gc_pause_seconds_sum` spikes; app threads freeze.
Diagnosis & Fix: Correlate heap usage with GC pauses. Switch to low-latency GC (G1GC/ZGC) and tune `-Xmx` heap size.

39. Cascading Latency Across Microservices — APM-039

Symptoms: Compounding delay in trace: Service A -> Service B -> Service C.
Diagnosis & Fix: Trace dependency graph to locate slow root service. Implement circuit breakers and enforce strict HTTP timeouts.

40. Message Queue Backlog Surge — APM-040

Symptoms: Kafka lag / RabbitMQ queue size grows exponentially.
Diagnosis & Fix: Compare consumption rate vs incoming rate. Autoscale consumer deployment replicas via KEDA based on lag metrics.

41. APM Trace Context Drop — APM-041

Symptoms: Fragmented, unlinked trace spans across microservices.
Diagnosis & Fix: Check W3C `traceparent` header propagation. Update app client libraries to forward tracing headers.

42. Thread Pool Task Rejections — APM-042

Symptoms: `thread_pool_rejected_tasks_total > 0`.
Diagnosis & Fix: App returns 503 errors under load. Expand worker thread pool sizes or scale out pod count horizontally.

43. Node.js / Python Memory Leak — APM-043

Symptoms: Runtime heap usage climbs steadily until OOMKilled.
Diagnosis & Fix: Profile heap snapshots over lifetime. Unbind global event listeners and fix unclosed file descriptors.

44. External Third-Party API Timeout — APM-044

Symptoms: Outbound HTTP call duration times out on external endpoint.
Diagnosis & Fix: Filter APM dependency map for external errors. Lower client timeouts, set fallback responses, and add caching.

SECTION 5: INGRESS, NETWORK & STORAGE (45 – 50)

45. Ingress Controller HTTP 502 Bad Gateway — NET-045

Symptoms: Ingress returns 502 status code to external clients.
Diagnosis & Fix: Check target endpoints (`kubectl get ep`). Align Service targetPort with container app port and fix failing probes.